



Modelling Financial Markets

Seminar 2: Market Efficiency — Hands-on



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Acronyms used in this material

- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BET-TR: BET Total Return (index)
- ▣ BIST: Borsa İstanbul (BIST 100 index) (the Istanbul Stock Exchange (BIST 100 index))
- ▣ BUX: Budapesti Értéktőzsde index (the Budapest Stock Exchange index)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CD: Chow–Denning (multiple variance-ratio test)
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ DFA: Detrended Fluctuation Analysis
- ▣ FRED: Federal Reserve Economic Data
- ▣ FTSE: Financial Times Stock Exchange (FTSE Russell, index provider)
- ▣ FWER: Family-Wise Error Rate
- ▣ GMM: Generalized Method of Moments
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ IPC: Índice de Precios y Cotizaciones (the Mexican Stock Exchange index)
- ▣ MSPE: Mean Squared Prediction Error
- ▣ OLS: Ordinary Least Squares
- ▣ PX: Prague Stock Exchange index (index Burzy cenných papírů Praha)
- ▣ RW1: Random Walk 1 (i.i.d. increments)
- ▣ RW3: Random Walk 3 (uncorrelated increments)
- ▣ SE: Standard Error
- ▣ TOM: Turn Of the Month
- ▣ US: United States
- ▣ VR: Variance Ratio
- ▣ WIG20: Warszawski Indeks Giełdowy 20 (the Warsaw Stock Exchange index of the 20 largest companies)

Outline

Introduction

Key Definitions

Part A: Pen and Paper

Part B: Computation and Inference

Part C: Open Question

Wrap-up

References

Seminar Objectives

By the end of this seminar you can

- ▣ Derive the null distribution of the variance ratio and the Stambaugh bias; compute runs tests and Hurst exponents step by step
- ▣ Test the random walk with inference robust to heteroskedasticity
- ▣ Measure time-varying efficiency with rolling windows and bootstrap bands
- ▣ Test calendar anomalies with HAC errors and multiple-testing corrections (Holm, StepM)
- ▣ Evaluate return predictability out of sample and event studies with clustered event days

Structure

- ▣ **Part A:** six pen-and-paper exercises and derivations (A1–A6)
- ▣ **Part B:** nine computational exercises with inference (B1–B9)
- ▣ **Part C:** one open question, a seed for the team project (C1), and an AI answer to critique (C2)

- ▣ Background: Lecture 2; textbooks [Campbell, Lo & MacKinlay \(1997\)](#), Ch. 2 and [Franke, Härdle & Hafner \(2019\)](#)

How to Work in This Seminar

- **[Solved]** problems show the full worked solution; **[Proposed]** problems show only the task: their solutions are discussed in class
- Every proposed problem names its **model**: the solved problem that uses the same technique
 - ▶ Every Part B exercise ends with interpretation questions: ask what would change the conclusion
- Report every statistic with its uncertainty (standard error, band or p -value)
- Always compare a classical test with its robust version: when do they disagree, and why?

Solved and Proposed Problems

Solved: worked examples in this deck

- Part A: A1 (variance ratio, step by step), A3 (runs test, step by step), A5 (Hurst exponent from a variance ratio)
- Part B: B1 (homoskedastic vs robust VR), B3 (rolling DFA with a wild-bootstrap null), B5 (turn of the month with HAC errors)

Proposed: you solve them

- Part A: A2 (model: A1), A4 (model: A1, A5), A6 (model: A5)
- Part B: B2 (model: B1), B4 (model: B3, B1), B6 (model: B5), B7 (model: B1), B8 (model: B5, A4), B9 (model: Seminar 0 Ex. 7, B5)
- Part C: C1, open question and project seed; C2, critique an AI answer (model: A1, B1, B7)

Setup: Python Environment and Data

-  **Open in Colab:** `chapter2_seminar_notebook.ipynb`
- Google Colab (recommended) or local Python ≥ 3.10 with `numpy`, `pandas`, `scipy`, `statsmodels`, `matplotlib`
- Series: S&P 500, BET, BET-TR, WIG20, BUX, PX, BIST 100, Bovespa, IPC, Nifty 50, Shanghai, Euro Stoxx 50, Bitcoin; daily closes up to 18 September 2026
- Equity indices: weekdays only, days with an unchanged close (holidays) removed; crypto keeps all 7 days of the week
- Returns: daily log returns $r_t = \ln P_t - \ln P_{t-1}$, each series on its own calendar

Key Definitions (1/3): Random Walk Tests

- **RW3**: uncorrelated increments, volatility may change; the hypothesis tested today
- **Variance ratio** Lo & MacKinlay (1988): $VR(q) = \text{Var}(r_t + \dots + r_{t-q+1}) / (q\text{Var}(r_t)) = 1 + 2 \sum_{k=1}^{q-1} (1 - k/q) \rho_k$
 - ▶ $z(q)$: assumes constant variance; $z^*(q)$: robust to heteroskedasticity (formulas in Lecture 2)
- **Chow–Denning** Chow & Denning (1993): $\max_i |z^*(q_i)|$ over m horizons; 5% critical value 2.49 for $m = 4$
- **Runs test** Wald & Wolfowitz (1940): number of runs of equal signs R , $\mathbb{E}[R] = 2n_1n_2/n + 1$, $\text{Var}[R] = 2n_1n_2(2n_1n_2 - n)/(n^2(n - 1))$
- **Hurst exponent** H : scaling of the range, $0.5 = \text{no memory}$; DFA exponent Peng et al. (1994); self-similar case $VR(q) = q^{2H-1}$

Key Definitions (2/3): Inference Tools

- ▣ **Robust SE of $\hat{\rho}_k$:** $\sqrt{\hat{\tau}_k}$, $\hat{\tau}_k = \sum_t e_t^2 e_{t-k}^2 / (\sum_t e_t^2)^2$, with $e_t = r_t - \bar{r}$
- ▣ **HAC** (Heteroskedasticity and Autocorrelation Consistent) errors [Newey & West \(1987\)](#): valid when residuals are heteroskedastic and autocorrelated
- ▣ **Wild bootstrap** [Kim \(2006\)](#): multiply centred returns by random signs ± 1 ; keeps the volatility path, destroys autocorrelation
- ▣ **Block bootstrap** [Künsch \(1989\)](#): resample blocks of consecutive days; keeps short-run dependence
- ▣ **FWER**: probability of at least one false rejection; **Bonferroni**: $p_i < \alpha/M$; **Holm**: sorted $p_{(j)} < \alpha/(M - j + 1)$, stop at the first failure

Key Definitions (3/3): Predictability and Event Studies

- **Stambaugh bias** Stambaugh (1999): $\mathbb{E}[\hat{\beta} - \beta] \approx -(\sigma_{uv}/\sigma_v^2)(1 + 3\rho)/T$ for a persistent predictor $x_{t+1} = \theta + \rho x_t + v_{t+1}$
- **Out-of-sample R^2** Welch & Goyal (2008): $R_{OS}^2 = 1 - \sum (r_t - \hat{r}_t)^2 / \sum (r_t - \bar{r}_t)^2$, $\bar{r}_t$ the recursive historical mean
- **Clark–West** Clark & West (2007): t -test on $f_t = (r_t - \bar{r}_t)^2 - [(r_t - \hat{r}_t)^2 - (\bar{r}_t - \hat{r}_t)^2]$ for nested models
- **Prediction-error variance** MacKinlay (1997): $\sigma_\varepsilon^2[1 + 1/T_0 + (r_{m,\tau} - \hat{\mu}_m)^2/((T_0 - 1)\hat{\sigma}_m^2)]$
- **StepM** Romano & Wolf (2005): step-down multiple testing with the joint bootstrap distribution of all statistics

A1: Variance Ratio, Step by Step [Solved]

Data and assumptions

- Eight daily returns in %, in time order: 1.0, -0.5, 0.8, 0.3, -1.2, 0.6, 0.9, -0.4; no data file is needed.
- Use the simple estimator without bias corrections: divide sums of squares by the number of terms.

Tasks

- (a) Compute the mean $\hat{\mu}$ and the one-day variance $\hat{\sigma}_1^2 = \frac{1}{T} \sum_t (r_t - \hat{\mu})^2$ with $T = 8$.
- (b) Form the seven overlapping two-day sums $s_t = r_t + r_{t-1}$ and compute $\hat{\sigma}_2^2 = \frac{1}{7} \sum_t (s_t - 2\hat{\mu})^2$.
- (c) Compute $\widehat{VR}(2) = \hat{\sigma}_2^2 / (2\hat{\sigma}_1^2)$, the lag-1 autocorrelation $\hat{\rho}_1$, and compare $\widehat{VR}(2)$ with $1 + \hat{\rho}_1$.

Report and interpret

- Report $\hat{\mu}$, $\hat{\sigma}_1^2$, $\hat{\sigma}_2^2$, $\widehat{VR}(2)$ and $\hat{\rho}_1$ with three decimals; say whether the series shows momentum or mean reversion, and why eight observations cannot support a test.

A1: Solution [Solved]

Step by step

- Mean: $\hat{\mu} = (1.0 - 0.5 + 0.8 + 0.3 - 1.2 + 0.6 + 0.9 - 0.4)/8 = 1.5/8 = 0.1875\%$.
- One-day variance: subtract $\hat{\mu}$ from each return, square, average over 8 terms: $\hat{\sigma}_1^2 = 0.5586$.
- Two-day sums: 0.5, 0.3, 1.1, -0.9, -0.6, 1.5, 0.5; subtract $2\hat{\mu} = 0.375$, square and average over 7 terms: $\hat{\sigma}_2^2 = 0.6292$.
- Variance ratio: $\widehat{VR}(2) = 0.6292/(2 \times 0.5586) = 0.563$.
- Autocorrelation: $\hat{\rho}_1 = \sum_t e_t e_{t-1} / \sum_t e_t^2 = -0.395$, so $1 + \hat{\rho}_1 = 0.605$; the gap to 0.563 comes from the end terms that a sample of eight cannot average out.

Interpretation

- $\widehat{VR}(2) < 1$ and $\hat{\rho}_1 < 0$: consecutive returns tend to reverse, i.e. mean reversion. With $T = 8$ the standard error of $\hat{\rho}_1$ is about $1/\sqrt{8} = 0.35$, so the value is not significant: the exercise trains the mechanics, not the inference.

A2: Derive the Variance Ratio and Its Null Distribution (1/2) [Proposed]

Solution discussed in class

Data and assumptions

- ▣ Returns r_t are covariance-stationary with variance γ_0 and autocorrelations ρ_k . Under RW1 (i.i.d.) the sample autocorrelations satisfy $\sqrt{T}(\hat{\rho}_1, \dots, \hat{\rho}_{q-1}) \rightarrow N(0, I)$; under RW3 the covariance is $\text{diag}(\delta_k)$.
- ▣ Numerical check: S&P 500, daily log returns 2000–2026, $T = 6\,714$, $\widehat{\text{VR}}(2) = 0.901$ (Lecture 2).

Tasks

- ▣ (a) Expand $\text{Var}(\sum_{j=0}^{q-1} r_{t-j})$ in autocovariances and derive $\text{VR}(q) = 1 + 2 \sum_{k=1}^{q-1} (1 - k/q) \rho_k$.
- ▣ (b) Apply the delta method to show that under RW1 $\sqrt{T}(\widehat{\text{VR}}(q) - 1) \rightarrow N(0, \frac{2(2q-1)(q-1)}{3q})$, and evaluate the variance for $q = 2, 5, 20$.

A2: Derive the Variance Ratio and Its Null Distribution (2/2) [Proposed]

Solution discussed in class

Tasks

- (c) Replace I by $\text{diag}(\delta_k)$ and write the asymptotic variance $\theta(q)$ used by the robust $z^*(q)$.
- (d) Compute $z(2)$ for the S&P 500 from the numbers above and decide at $\alpha = 5\%$.

Report and interpret

- Report the derivation in three lines, the three variances and $z(2)$; explain why the same weights $1 - k/q$ appear in the robust version. Model: A1.

A3: Runs Test, Step by Step [Solved]

Data and assumptions

- Signs of 20 consecutive daily returns: ++-+++--+-++---+---; zero returns do not occur. Under H_0 the signs are independent with a constant probability of a positive day [Wald & Wolfowitz \(1940\)](#).

Tasks

- (a) Count the positive signs n_1 , the negative signs n_2 and the runs R (maximal blocks of equal signs).
- (b) Compute $\mathbb{E}[R] = 2n_1n_2/n + 1$ and $\text{Var}[R] = 2n_1n_2(2n_1n_2 - n)/(n^2(n - 1))$ with $n = n_1 + n_2$.
- (c) Compute $z = (R - \mathbb{E}[R])/\sqrt{\text{Var}[R]}$, the two-sided p -value from the Normal distribution, and decide at $\alpha = 5\%$.

Report and interpret

- Report n_1 , n_2 , R , $\mathbb{E}[R]$, $\text{Var}[R]$, z and p ; say what too few or too many runs would mean for the sign of the autocorrelation.

A3: Solution [Solved]

Step by step

- (a) Counting: $n_1 = 12$ plus signs, $n_2 = 8$ minus signs, $n = 20$; the sign changes 10 times, so $R = 11$ runs.
- (b) $\mathbb{E}[R] = 2 \cdot 12 \cdot 8/20 + 1 = 10.60$; $2n_1n_2 = 192$ and $\text{Var}[R] = 192 \cdot (192 - 20)/(400 \cdot 19) = 4.345$, so $\text{SD}[R] = 2.085$.
- (c) $z = (11 - 10.60)/2.085 = 0.40/2.085 = 0.19$; two-sided $p = 0.85$.

Interpretation

- $p = 0.85 > 0.05$: do not reject independence; the observed number of runs is what chance produces.
- Too few runs ($z < 0$) would mean persistent signs (positive autocorrelation); too many ($z > 0$), alternating signs (reversal). With $n = 20$ the Normal approximation is rough: exact tables are preferable.

A4: The Stambaugh Bias [Proposed]

Solution discussed in class

Data and assumptions

- Predictive system: $r_{t+1} = \alpha + \beta x_t + u_{t+1}$ and $x_{t+1} = \theta + \rho x_t + v_{t+1}$; write $u = \phi v + e$ with $\phi = \sigma_{uv}/\sigma_v^2$ and $\mathbb{E}[e | x] = 0$; Kendall: $\mathbb{E}[\hat{\rho} - \rho] \approx -(1 + 3\rho)/T$ **Stambaugh (1999)**.
- Illustrative annual inputs: $\rho = 0.98$, $T = 60$ years, $\text{corr}(u, v) = -0.9$, $\sigma_u = 0.17$, $\sigma_v = 0.14$.

Tasks

- (a) Show that $\hat{\beta} - \beta = \phi(\hat{\rho} - \rho) + \sum \tilde{x}_t e_{t+1} / \sum \tilde{x}_t^2$, where $\tilde{x}$ is the demeaned predictor, and derive $\mathbb{E}[\hat{\beta} - \beta]$.
- (b) Compute ϕ , $\mathbb{E}[\hat{\rho} - \rho]$ and the bias of $\hat{\beta}$ with the inputs above.
- (c) Compute $\text{SE}(\hat{\beta}) \approx \sigma_u / (\sigma_x \sqrt{T})$ with $\sigma_x = \sigma_v / \sqrt{1 - \rho^2}$ and express the bias in standard errors.

Report and interpret

- Report ϕ , the bias, the SE and their ratio; explain why a persistent predictor with correlated innovations makes “significant” predictability too easy. Model: A1, A5.

A5: Hurst Exponent from a Variance Ratio [Solved]

Data and assumptions

- For exactly self-similar increments (fractional Gaussian noise) $VR(q) = q^{2H-1}$. Data: BET, daily log returns 2000–2026 (data/market/BET.csv, close), $\widehat{VR}(20) = 1.377$ (Lecture 2).

Tasks

- (a) Take logarithms of $VR(q) = q^{2H-1}$ and solve for H .
- (b) Compute $\hat{H}$ for BET with $q = 20$.
- (c) Compare it with the Anis–Lloyd corrected R/S estimate 0.533 of Lecture 2.

Report and interpret

- Report the formula and $\hat{H}$ with three decimals; say whether this value proves long memory.

A5: Solution [Solved]

Step by step

- (a) $\ln VR(q) = (2H - 1) \ln q$, hence $H = 0.5 + \ln VR(q)/(2 \ln q)$.
- (b) $\ln 1.377 = 0.3199$ and $2 \ln 20 = 5.991$, so $\hat{H} = 0.5 + 0.3199/5.991 = 0.553$.
- (c) The corrected R/S gives 0.533: both estimates lie above 0.5 and point to mild persistence.

Interpretation

- The formula assumes exact self-similarity. An AR(1) with $\rho_1 = 0.109$ (short memory) also gives $VR(20) > 1$ and hence $\hat{H} > 0.5$: this value alone does not prove long memory (see A6).

A6: Hurst Exponent of the S&P 500 [Proposed]

Solution discussed in class

Data and assumptions

- S&P 500, daily log returns 2000–2026 (data/market/GSPC.INDX.csv, close), $\widehat{VR}(20) = 0.753$; a short-memory benchmark: AR(1) with $\rho_1 = -0.1$, so $\rho_k = (-0.1)^k$.

Tasks

- (a) Compute $\hat{H}$ with the formula of A5 and say whether the index is persistent or anti-persistent.
- (b) For the AR(1), compute $VR(20) = 1 + 2 \sum_{k=1}^{19} (1 - k/20) \rho_k$ and the implied $H(20)$.
- (c) Show that for any short-memory process ($\sum_k |\rho_k| < \infty$) $VR(q) \rightarrow 1 + 2 \sum_{k \geq 1} \rho_k = 2\pi f(0)/\gamma_0$, hence $H(q) \rightarrow 0.5$; check with $q = 250$.

Report and interpret

- Report $\hat{H}$, $VR(20)$, $H(20)$, $VR(250)$, $H(250)$; conclude whether $H < 0.5$ needs long memory. Model: A5.

B1: Homoskedastic vs Robust Variance Ratio [Solved]

Data and assumptions

- S&P 500 (GSPC.INDX) and BET (BET), daily closes 2000–2026 from data/market; weekdays only, days with an unchanged close removed; daily log returns, $T = 6\,714$ and $6\,670$.
- H_0 for $z(q)$: RW1 (i.i.d.); H_0 for $z^*(q)$: RW3 (uncorrelated, heteroskedasticity allowed) Lo & MacKinlay (1988).

Tasks

- (a) Compute $\widehat{VR}(q)$ with the Lo–MacKinlay bias corrections for $q = 2, 5, 20$.
- (b) Compute the homoskedastic $z(q)$ and the robust $z^*(q)$ with their two-sided p -values and decide at $\alpha = 5\%$.
- (c) Run the Chow–Denning joint test over $q = 2, 5, 10, 20$ (critical value 2.49 at 5%) Chow & Denning (1993).

Report and interpret

- Report a table of $\widehat{VR}$, z , z^* and the CD statistic with its p -value; explain why $|z| > |z^*|$, whether the sign of $VR - 1$ agrees with $\hat{\rho}_1$, and whether you would trade the S&P 500 reversal.

B1: Solution (1/2) [Solved]

Step by step

	$\widehat{VR}(2)$	$z(2)$	$z^*(2)$	$\widehat{VR}(5)$	$z(5)$	$z^*(5)$	$\widehat{VR}(20)$	$z^*(20)$
S&P 500	0.901	-8.12	-3.64	0.830	-6.37	-2.77	0.753	-1.86
BET	1.108	8.83	3.84	1.173	6.44	2.98	1.377	3.31

- (a)–(b) Steps: overlapping q -day sums, $\hat{\sigma}_c^2(q)/\hat{\sigma}_a^2$, then $z = (\widehat{VR} - 1)/\sqrt{2(2q - 1)(q - 1)/(3qT)}$ and $z^* = \sqrt{T}(\widehat{VR} - 1)/\sqrt{\hat{\theta}(q)}$.
- S&P 500: $z^*(2) = -3.64$ ($p = 0.0003$), $z^*(5) = -2.77$ ($p = 0.0056$): reject RW3; $z^*(20) = -1.86$ ($p = 0.063$): do not reject.
- BET: $z^*(2) = 3.84$ ($p = 0.0001$), $z^*(20) = 3.31$ ($p = 0.0009$): reject at every horizon.

B1: Solution (2/2) [Solved]

Step by step

- (c) Chow–Denning = $\max_q |z^*(q)|$: S&P 500 3.64 ($p = 0.0011$), BET 3.84 ($p = 0.0005$); both exceed 2.49, so both reject RW3 jointly at 5%.

Interpretation

- $|z|$ is two to three times $|z^*|$: volatility clustering makes $\text{Var}(\hat{\rho}_k) > 1/T$, so the homoskedastic test overstates the evidence.
- Signs agree with $\hat{\rho}_1$: VR < 1 (reversal) for the S&P 500, VR > 1 (persistence) for BET, growing with q .
- A statistical rejection is not a trading rule: trading costs, risk and stability over time (B3, B7) are missing from the test.

B2: BET-TR and Emerging Markets [Proposed]

Solution discussed in class

Data and assumptions

- BET-TR (BETTR.INDX, 2014–2026) and the indices WIG20, BUX, PX, BIST 100, Bovespa, IPC, Nifty 50, Shanghai (2000–2026), daily closes from data/market, each on its own calendar; daily log returns.

Tasks

- (a) For each series compute $\widehat{VR}(2)$, $z(2)$, $z^*(2)$ and $z^*(5)$ as in B1.
- (b) Compute the Chow–Denning statistic over $q = 2, 5, 10, 20$ and its p -value.
- (c) Count the rejections at $\alpha = 5\%$ with $z(2)$ and with Chow–Denning.

Report and interpret

- Report one table (one row per market); explain the gap between the two counts, why BET-TR can differ from BET (dividends, sample), and whether emerging markets are less efficient as a group. Model: B1.

B3: Rolling DFA on the S&P 500 (1/2) [Solved]

Data and assumptions

- S&P 500 daily log returns 2000–2026 (GSPC.INDX); rolling windows of 500 trading days moved by 21 days; DFA exponent α_{DFA} (linear detrending) with box sizes from 10 to $T/4$ [Peng et al. \(1994\)](#).
- H_0 : RW3, i.e. no serial correlation, volatility clustering allowed.

Tasks

- (a) Compute α_{DFA} in each window.
- (b) In each window build the null band by wild bootstrap [Kim \(2006\)](#): 199 series $e_t \eta_t$, $\eta_t = \pm 1$ with probability 1/2, e_t the demeaned returns; keep the 2.5% and 97.5% quantiles.

B3: Rolling DFA on the S&P 500 (2/2) [Solved]

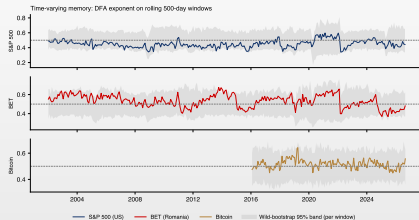
Tasks

- (c) Compute the share of windows outside their band and compare it with one fixed i.i.d. Student- t_4 band (300 simulations of length 500).

Report and interpret

- Report the chart, both shares, the mean, minimum and maximum exponent with dates; explain why the null must keep the volatility path and why the band is wider than the full-sample one.

B3: Solution (1/2) [Solved]



Step by step

- (a)–(b) 296 windows; the per-window band averages $[0.359, 0.644]$ and its width ranges from 0.208 to 0.500 (widest in the window ending 2022-02-11).

B3: Solution (2/2) [Solved]

Step by step

- (c) Outside their own band: 2.0% of windows, below the 5% expected by chance; with the fixed Student- t_4 band $[0.378, 0.614]$: 6.4%.
- Mean exponent 0.450; minimum 0.341 (window ending 2022-03-15); maximum 0.598 (2021-02-12); both extremes are in windows that contain the 2020 crash.

Interpretation

- No evidence of memory beyond chance: the mean below 0.5 matches the negative $\hat{\rho}_1$ (mild reversal). RW3 allows volatility clustering, which raises α_{DFA} ; sign flips keep $|e_t|$ and remove serial correlation, so the band adapts to each window.
- 500 observations instead of 6 714: the sampling error of a scaling exponent grows roughly like $1/\sqrt{T}$, hence the wide band.

B4: Adaptive Efficiency of Bitcoin [Proposed]

Solution discussed in class

Data and assumptions

- Bitcoin (BTC-USD.CC, close), daily log returns from 2011, 7 days a week; rolling windows of 365 days moved by 30 days; H_0 : RW3 Lo (2004).

Tasks

- (a) Compute the robust $z^*(5)$ of the Lo–MacKinlay test in each window.
- (b) In each window build a wild-bootstrap band for $z^*(5)$ Kim (2006): 99 replications of $e_t \eta_t$, $\eta_t = \pm 1$; keep the 2.5% and 97.5% quantiles.
- (c) Count the windows outside their band and outside ± 1.96 ; compare with the chance level $0.05 \times (\text{number of windows})$ and with Urquhart (2016).

Report and interpret

- Report the chart, the rejection dates with z^* and band, and both shares; say whether Bitcoin becomes more efficient over time and why the band is not ± 1.96 . Model: B3, B1.

B5: The Turn-of-the-Month Effect with HAC Errors [Solved]

Data and assumptions

- S&P 500 and BET, daily log returns 2000–2026 in basis points (bp); $\text{TOM}_t = 1$ on the last trading day of a month and the first three trading days of the next [Ariel \(1987\)](#), 0 otherwise.

Tasks

- (a) Estimate $r_t = a + b \text{TOM}_t + u_t$ by OLS; b is the extra return on TOM days.
- (b) Compute the OLS and the Newey–West HAC standard errors of $\hat{b}$ with 5 lags [Newey & West \(1987\)](#), the two t -statistics and the HAC p -value; decide at $\alpha = 5\%$.
- (c) Convert $\hat{b}$ into an extra return per month (4 TOM days).

Report and interpret

- Report a table (share of TOM days, $\hat{a}$, $\hat{b}$, both SEs and t); explain why OLS and HAC errors differ and which explanations (flows, pension funds, rebalancing) you would test next.

B5: Solution (1/2) [Solved]

Result

	TOM share	$\hat{a}$ (bp)	$\hat{b}$ (bp)	SE_{OLS}	t_{OLS}	SE_{HAC}	t_{HAC} (p)
S&P 500	19.1%	1.8	3.3	3.8	0.88	3.5	0.95 (0.34)
BET	19.3%	3.0	17.8	4.3	4.11	4.8	3.74 (0.0002)

Step by step

- (a) $\hat{a}$ is the mean on other days, $\hat{a} + \hat{b}$ the mean on TOM days: S&P 500 5.2 vs 1.8 bp; BET 20.8 vs 3.0 bp.
- (b) BET: $t_{HAC} = 17.8/4.8 = 3.74$, $p = 0.0002$: reject $b = 0$. S&P 500: $t_{HAC} = 0.95$, $p = 0.34$: do not reject.

B5: Solution (2/2) [Solved]

Step by step

- (c) BET: $4 \times 17.8 \approx 71$ bp extra per month; S&P 500: $4 \times 3.3 \approx 13$ bp, not distinguishable from zero.

Interpretation

- For BET the HAC error exceeds the OLS error (volatility clustering and $\rho_1 > 0$), so t falls from 4.11 to 3.74; the effect survives and is economically large.
- One market and one effect were chosen among many: before believing it, correct for the number of effects tried (B6), then look for a mechanism such as salary and pension inflows at the start of the month.

B6: Fifteen Calendar Tests and Multiple Testing [Proposed]

Solution discussed in class

Data and assumptions

- Five markets (S&P 500, BET, Euro Stoxx 50, WIG20, BUX), daily log returns 2000–2026, each on its own calendar; three effects: day of the week (Wald test of equal means), January (monthly returns, HAC with 3 lags), turn of the month (daily, HAC with 5 lags).

Tasks

- (a) Compute the 15 p -values and the raw decisions at $\alpha = 5\%$.
- (b) Apply Bonferroni ($p_i < \alpha/15$) and Holm (sorted $p_{(j)} < \alpha/(16 - j)$), stop at the first failure).
- (c) Apply Romano–Wolf StepM [Romano & Wolf \(2005\)](#): resample blocks of 3 calendar months, the same for all markets, 999 times; step down on the minimum bootstrap p -value of the hypotheses not yet rejected.

Report and interpret

- Report the table of p -values with the three decisions, the FWER of 15 independent tests, and which effects pass $|t| > 3$ [Harvey, Liu & Zhu \(2016\)](#); say whether StepM buys power over Holm here, and why. Model: B5.

B7: Is the S&P 500 Reversal a Crisis Effect? [Proposed]

Solution discussed in class

Data and assumptions

- S&P 500 daily log returns 2000–2026 ($T = 6\,714$); crisis days: 1 Sep 2008 – 30 Jun 2009 and 15 Feb – 30 Jun 2020.

Tasks

- (a) Compute $\hat{\rho}_1$ with the i.i.d. $t = \hat{\rho}_1 \sqrt{T}$ and the robust $t = \hat{\rho}_1 / \sqrt{\hat{\tau}_1}$ on all days.
- (b) Remove the crisis days and repeat (a); report the share of days removed.
- (c) Compute $z^*(5)$ in both samples and decide at $\alpha = 5\%$.

Report and interpret

- Report a two-row table (T , $\hat{\rho}_1$, both t , $z^*(5)$); say how much of the autocorrelation comes from the crises, why the i.i.d.–robust gap shrinks, and when removing crisis days becomes data snooping. Model: B1.

B8: Predictive Regression and an Out-of-Sample Test (1/2) [Proposed]

Solution discussed in class

Data and assumptions

- S&P 500 monthly log excess return (dividends included, minus the 3-month T-bill rate, FRED TB3MS), 1934–2026, on the lagged log dividend–price ratio $dp_t = \ln D_t - \ln P_t$ [Stambaugh \(1999\)](#).
- Prices and dividends: Shiller's monthly S&P Composite (to 2026-06), extended with S&P 500 closes (GSPC.INDX) and SPY dividends (SPY.US).

Tasks

- (a) Estimate $r_{t+1} = \alpha + \beta dp_t + u_{t+1}$ by OLS with Newey–West errors (12 lags); estimate the AR(1) of dp and $\text{corr}(\hat{u}, \hat{v})$; correct $\hat{\beta}$ for the Stambaugh bias (A4).
- (b) From January 1970 forecast recursively (expanding window) and compute R_{OS}^2 against the historical mean [Welch & Goyal \(2008\)](#).

B8: Predictive Regression and an Out-of-Sample Test (2/2) [Proposed]

Solution discussed in class

Tasks

- (c) Compute the Clark–West t (HAC, 12 lags, one-sided) Clark & West (2007) and apply the Campbell–Thompson restrictions Campbell & Thompson (2008).

Report and interpret

- Report $\hat{\beta}$, t , $\hat{\rho}$, the bias, R_{OS}^2 and the Clark–West t ; say whether the in-sample evidence survives, why Diebold–Mariano is wrong for nested models, and what $R^2 = 0.5\%$ is worth to a mean–variance investor. Model: B5, A4.

B9: The Bank Tax, TLV and BRD Jointly (1/2) [Proposed]

Solution discussed in class

Data and assumptions

- Banca Transilvania (TLV.R0) and BRD (BRD.R0), `adjusted_close`, and the BET index (BET, `close`), joined on common trading days; event: the bank tax announced on the evening of 18 December 2018, day 0 = 19 December (Seminar 0, Ex. 7).
- Market model $r_{i,t} = a_i + b_i r_{m,t} + \varepsilon_{i,t}$ estimated on days $[-250, -11]$ ($T_0 = 240$).

Tasks

- (a) For each bank compute AR_0 , the naive $t = AR_0 / \hat{\sigma}_\varepsilon$ and the t with the prediction-error variance $\hat{\sigma}_\varepsilon^2 [1 + 1/T_0 + (r_{m,0} - \hat{\mu}_m)^2 / ((T_0 - 1)\hat{\sigma}_m^2)]$ MacKinlay (1997).
- (b) For the mean AR_0 compute the t assuming independent banks and the t of the equal-weighted portfolio; compute the residual correlation $\bar{r}$ and the Kolari–Pynnönen factor $\sqrt{(1 - \bar{r}) / (1 + (N - 1)\bar{r})}$ Kolari & Pynnönen (2010).

B9: The Bank Tax, TLV and BRD Jointly (2/2) [Proposed]

Solution discussed in class

Tasks

- (c) Compute $CAR[0, +5]$ and its t with the full covariance of the prediction errors.

Report and interpret

- Report one table (TLV, BRD, portfolio); say how much of “ -8.3% , $t = -10$ ” survives, why a common event day calls for the portfolio or Kolari–Pynnönen test, and whether the reversal contradicts semi-strong efficiency. Model: Seminar 0 Ex. 7, B5.

C1: Did the BVB Become More Efficient after the 2020 FTSE Upgrade? (1/2) [Proposed]

Solution discussed in class

Data and assumptions

- **FTSE Russell (2019)**: Romania moved from Frontier to Secondary Emerging on 21 September 2020; more foreign investors and analysts should speed up price discovery. Background: **Pele & Mazurencu-Marinescu (2012)**; the question extends Seminar 1, C1.
- **Data**: BET (data/market/BET.csv, close), daily log returns five years before and five years after the event; WIG20 and BUX as a comparison group.

Tasks

- (a) Compute $\hat{\rho}_1$ and $z^*(5)$ in both five-year windows.
- (b) Build a moving-block bootstrap 95% interval for the change in $\hat{\rho}_1$ (blocks of 20 days, 999 replications) **Künsch (1989)**.

C1: Did the BVB Become More Efficient after the 2020 FTSE Upgrade? (2/2) [Proposed]

Solution discussed in class

Tasks

- (c) Add a rolling view (B3) and the same statistics for WIG20 and BUX to separate the event from common shocks; list the confounders (COVID-19, Hidroelectrica listed in 2023, liquidity).

Report and interpret

- Deliver a Quantlet folder in the team repository and one slide with the conclusion, the main limitation and the link with the Adaptive Markets Hypothesis [Lo \(2004\)](#).

C2: Critique an AI Answer (1/2) [Proposed]

Solution discussed in class

- **The prompt** a student sent to an AI (Artificial Intelligence) assistant: “Test whether the S&P 500 (daily closes, 2000–2026) follows a random walk with the variance ratio. Give Python code and interpret the result”
- **The answer:** the code below, followed by the interpretation

```
r = np.log(sp).diff().dropna()           # daily log returns
T = len(r)
def vr(q):
    rq = r.rolling(q).sum().dropna()      # overlapping q-day returns
    return rq.var() / (q * r.var())
q = 5
z = (vr(q) - 1) / np.sqrt(2 * (2*q - 1) * (q - 1) / (3 * q * T))
p = 2 * stats.norm.sf(abs(z))
```

- **Interpretation of the AI**
 - ▶ “VR(2) = 0.901, VR(5) = 0.830, $z(5) = -6.37$, $p < 0.001$: the random walk is rejected”
 - ▶ “Since VR(2) = $1 + 2\rho_1$, the first-order autocorrelation is $\rho_1 \approx -0.050$ ”
 - ▶ “VR < 1 means positive autocorrelation, i.e. momentum: buy the S&P 500 after up days”

C2: Critique an AI Answer (2/2) [Proposed]

Solution discussed in class

Task

- (a) Find the errors: there are two or three; the rest of the answer is correct
- (b) For each error, say how you would detect it: a test, a published number, the lecture or the library documentation
- (c) Give the corrected code and the corrected interpretation
- (d) Log each error as in `AI_ERRORS.md`: what was asked, what the AI answered, what was wrong, how you found out, the correct version
- Model: A1, B1 and B7

Why this exercise

- AI code usually runs without an error message: the mistakes are in the statistics, not in the syntax
- Course rule: you may use AI, declare it in `AI_USE.md` and log at least three errors you caught in `AI_ERRORS.md`

Common Pitfalls

- ▣ Using $\pm 1.96/\sqrt{T}$ bands or homoskedastic z with volatility clustering: too many rejections
- ▣ Reading the best of many horizons, markets or calendar effects without a multiple-testing correction
- ▣ Taking an uncorrected R/S slope above 0.5 as proof of long memory
- ▣ Counting overlapping rolling windows as independent evidence
- ▣ Equating statistical rejection with a profitable strategy: costs and risk are missing
- ▣ Filling holidays with the previous close: creates zero returns and fake runs

Take-Home Assignment (Team Project)

- ▣ Choose one market not covered in class (another index, an exchange rate or a crypto-asset)
- ▣ Run the full battery: robust $\hat{\rho}_k$, $\tilde{Q}(10)$, runs, $z^*(q)$, Chow–Denning, Lo's V , DFA with a wild-bootstrap band
- ▣ Add a rolling analysis with a proper null band and one calendar effect with HAC errors and Holm correction
- ▣ Deliver a Quantlet folder (code, Metainfo, charts) and three slides: evidence, interpretation, limitations
- ▣ Log every AI error you catch, in this seminar or in the project, in `AI_ERRORS.md`

Further Study on Quantinar

Advanced courses on quantinar.com

-  Efficiency of cryptocurrency markets - A GMM-based analysis
-  Generalized Method of Moments

Code and data

- Seminar notebook (Colab); Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_02

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